

hw 0413

11.3

(14)

1. $(-0.1438, -0.0242)$

沒包含 0 $\Rightarrow$ 有顯著差異

$\bar{X} < \bar{Y}$

2. $3-1 (-0.0178, 0.1018)$

No significantly difference

3. $3-2 (0.0662, 0.1858)$

$\bar{X} > \bar{Y}$

(16)

$q_{0.01, 4, 16} = 5.19$

$HSD = 5.19 \sqrt{\frac{41.35}{5}} = 14.91$

X_2 和其他都有顯著差異

X_3, X_4 有

X_1 和 X_3, X_4 沒

(17)

	df	SS	MS	F
treatment	2	9486.69	4743.35	4.47
error	29	25930	897.59	
total	29	300936.69		

p-value = 0.021 < 0.05, reject H_0

b. $\bar{X}_1 = 608 \quad \bar{X}_2 = 653$

95% C.I. $-(108.94, 38.94)$

c. $HSD = 101.43$

$\bar{A} = 608 \quad \bar{B} = 653 \quad \bar{C} = 532$

Physical science, social science 有顯著差異

④ a. Blocks: items
treatments: scores

k. ANOVA	SS	df	MS	F
treatment	13.1923	3	4.3974	26.52
block	40.9949	9	5.8514	
error	3.4819	21	0.1658	

$$P \approx 2.45 \times 10^{-9} < 0.05$$

reject H_0

$$c. F = \frac{5.8514}{0.1658} \approx 35.32$$

$$P\text{-value} \approx 7.16 \times 10^{-10} \rightarrow \text{顯著差異}$$

⑤ (a) $q_{0.05}(k, df) \rightarrow q_{0.05}(4, 21) \approx 3.94$

$$(b) W = q_{0.05}(4, 21) \sqrt{\frac{MSE}{b}} = \sqrt{\frac{0.1658}{8}} = 0.567$$

(c) Tukey

① Vons vs. Ralphs 不顯

② Vons vs. Starter Bros 不顯

③ Vons vs. WinCo 顯

④ Ralphs vs. Starter Bros 不

⑤ Ralphs vs. WinCo 顯

⑥ Starter Bros vs. WinCo 顯

⑥ (a) Treatments: 三種

Blocks: 5個 investigators

	SS	df	MS	F	P-value
treat	107.2	2(3-1)	53.6	11.65	0.00499
Block	176.4	4(5-1)	44.1	9.49	0.00452
E	38.8	8(3*5-1)	4.85		
T	322.4				

(b) $H_0: \mu_1 = \mu_2 = \mu_3$

H_a : 至少一个不同

$p = 0.00499 < 0.01 \Rightarrow \text{reject } H_0$

$\Rightarrow$ 当 $\alpha = 0.01$, 有 sufficient evidence that
3 mixture 不同

(c) $p = 0.00452 < 0.01$ 显著差异 $\Rightarrow$ blocking 有效

(d)

Mix 1 v.s Mix 3 显著不同